

SUBJECT	Lower Secondary Science Year 8
TOPIC	Electricity and Magnetism
ASSIGNMENT	Three
STUDENT NAME	

FEEDBACK AND REFLECTION:

- ☐ I have read the feedback on my last assignment carefully.
- ☐ I have considered whether any of that feedback is relevant to this assignment.

One thing I will try to do either the same or differently in this assignment as a result of my previous feedback is: _____

ASSIGNMENT GUIDANCE:

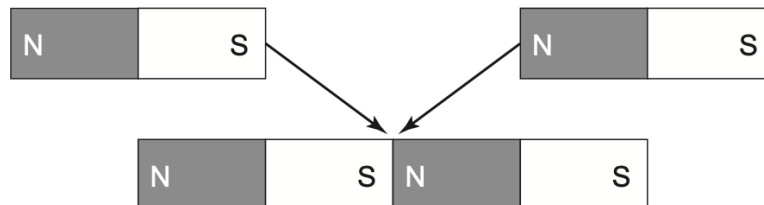
- You should complete this assignment by printing out this sheet and then handwriting in black ink, unless you have obtained specific permission to submit typed work. Then scan it in and submit it, as a single document.
- This is not a timed assignment. We estimate that this assignment may take you between 40 and 50 minutes for the assignment itself. However, if you need to take longer, then please do so.
- Make sure you write your full name clearly in the box at the top of this page.
- In order to complete this assignment, you will need a calculator, ruler, your coursebook and notes.

LS Science Assignment Three

ELECTRICITY AND MAGNETISM

1. Angelique puts two magnets near each other.

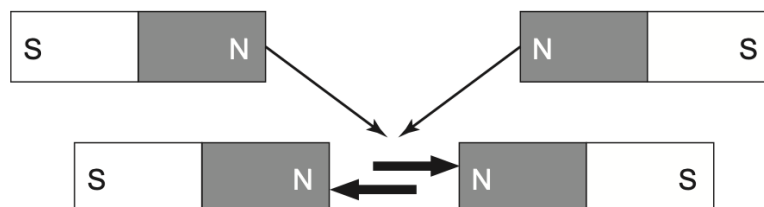
a. Look at what happens.



Explain what is happening in this diagram.

(2 marks)

b. Angelique puts the two magnets near each other again.



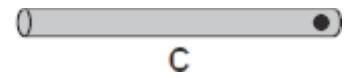
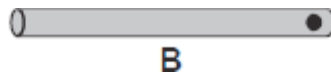
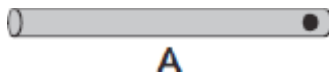
Explain what is happening in this diagram.

(2 marks)

(Total 4 marks)

2. Hannah has three rods (A, B and C) made from different metals.

One rod is a **magnet**; one is made of **copper**; and one is made of **iron**. She does not know which rod is which.



Each rod has a dot at one end.

a. Hannah uses **only** a bar magnet to identify each rod.

She puts each pole of the bar magnet next to the dotted end of each rod. Complete Hannah's observations in the table below.

Write if each rod is **copper**, **iron**, or a **magnet**.

test	observations	type of rod
 rod A	attract	Rod A is
 rod A	attract	_____
 rod B	nothing happens	Rod B is
 rod B	_____	_____
 rod C	attract	Rod C is
 rod C	_____	_____

(5 marks)

- b. Hannah uses the iron rod to make an electromagnet.

Give two ways Hannah could make the electromagnet stronger.

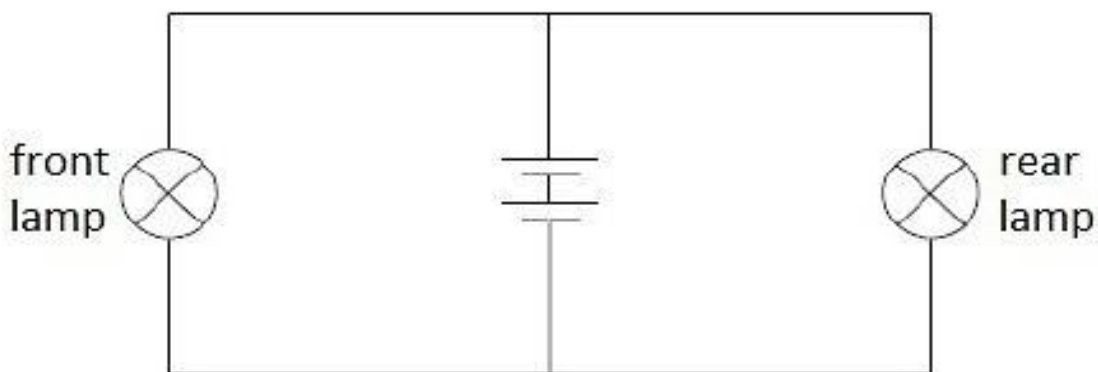
1. _____
2. _____

(2 marks)
(Total 7 marks)

3. Nina's bicycle has a front lamp and a rear lamp. Both lamps are connected to the same battery.



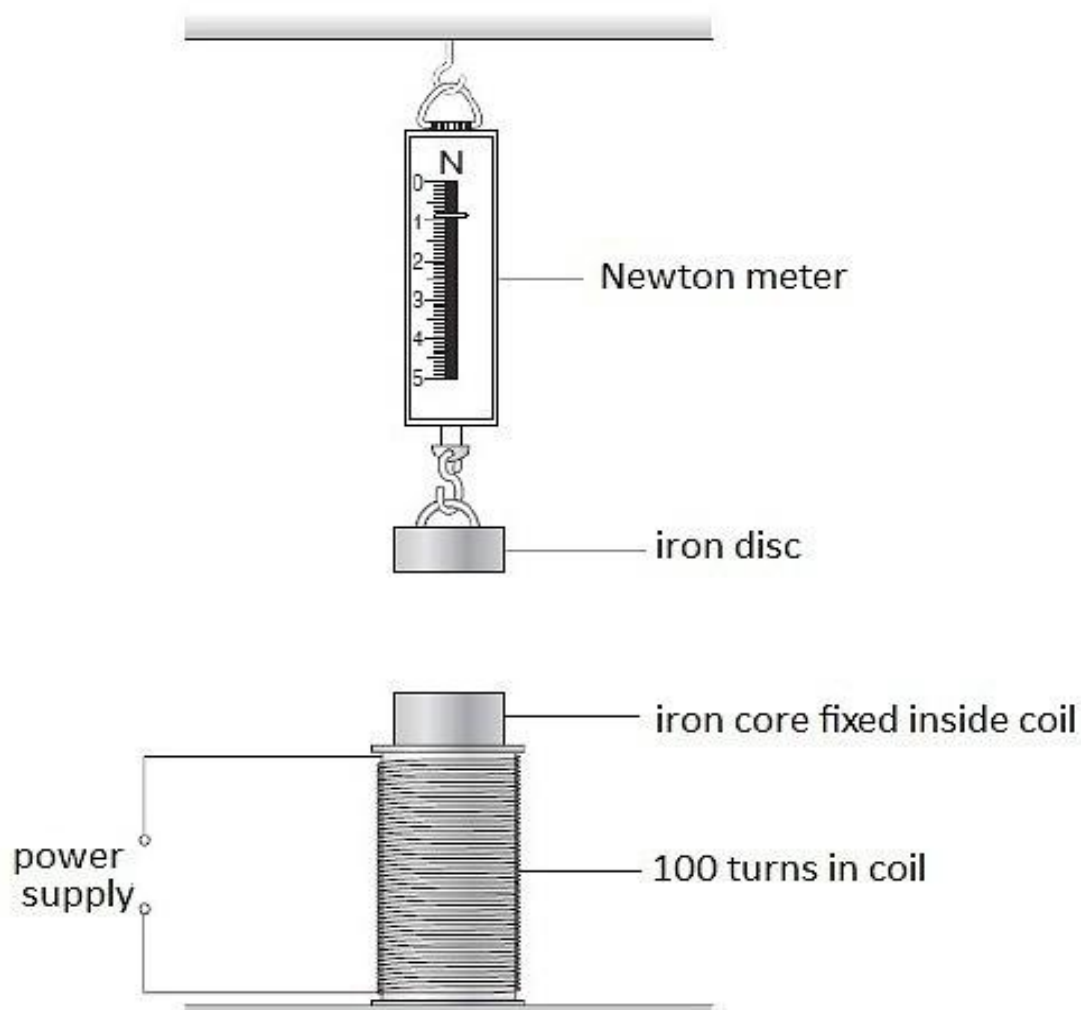
The circuit diagram for the lamps is drawn below.



- a. On the circuit diagram above, place a letter **A** to show the position of a switch to turn **only** the front lamp on and off
- b. On the circuit diagram above, place a letter **B** to show the position of a switch to turn **both** lamps on and off at the same time.

(Total 2 marks)

4. Mary used the apparatus below to test the strength of an electromagnet.



She used the reading on the Newton meter to measure the force of the magnet on the iron disc.

- a. Explain why this reading on the Newton meter increases when a current passes through the coil.

(2 marks)

- b. When a current passes through the coil, some of the electrical energy is changed to thermal energy.

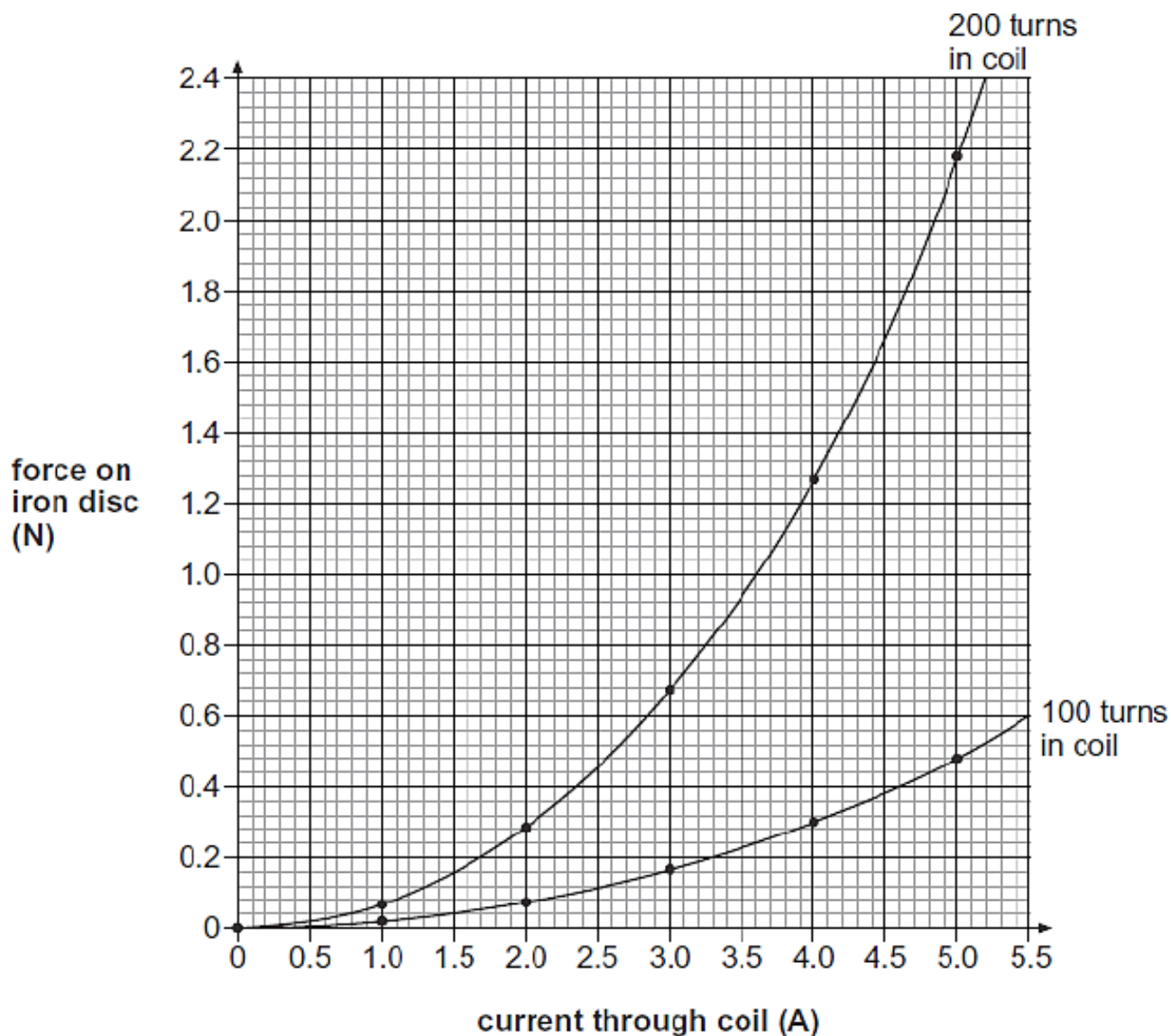
What would happen to the coil if the current passing through it was too large?

(1 mark)

- c. Mary made two electromagnets: one with 100 turns of wire in the coil, and one with 200 turns.

She varied the current through the coil of each electromagnet.

She measured the force of each electromagnet on the iron disc. The graph shows her results.

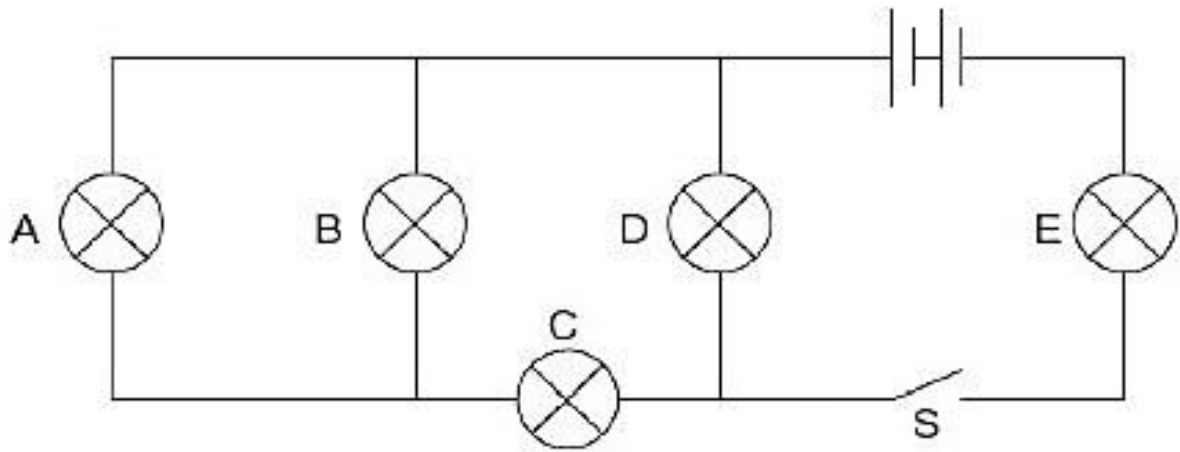


Write **two** conclusions that Mary could make from these results.

1. _____
- _____
2. _____
- _____

(2 marks)
(Total 5 marks)

5. a. Max built **circuit 1** as shown below.



circuit 1

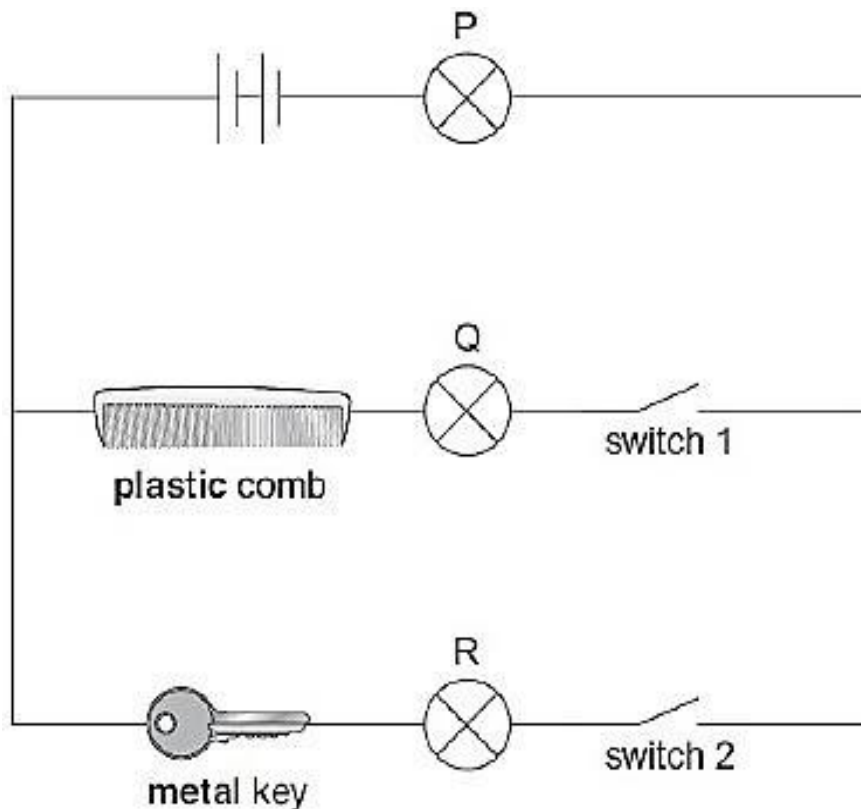
He closed the switch, S, and all the bulbs came on. One of the bulbs then broke and **all** the bulbs went off.

Which bulb must have broken? Give the letter.

(1 mark)

- b. Max built **circuit 2** as shown below.

He connected a plastic comb and a metal key in different parts of the circuit.



circuit 2

Look carefully at **circuit 2** on the previous page.

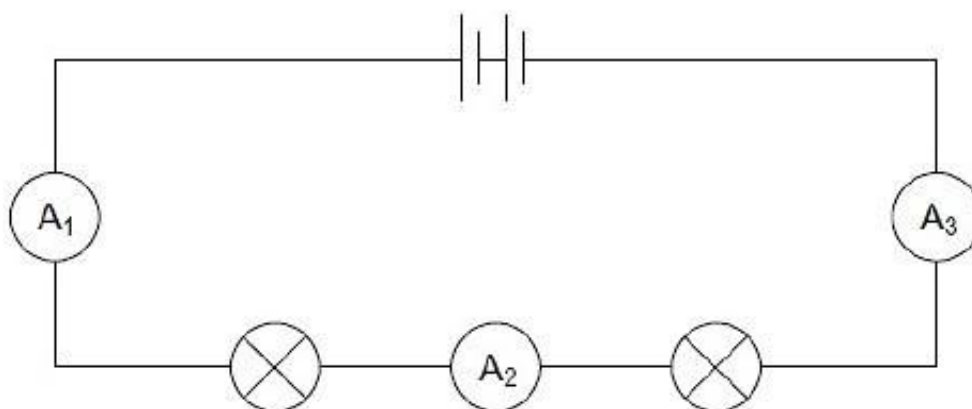
Complete the table below to show which bulbs in circuit 2 will be on or off when different switches are open or closed.

Write **on** or **off** in the boxes below.

switch 1	switch 2	bulb P	bulb Q	bulb R
open	open	off	off	off
open	closed			
closed	open			

(6 marks)

- c. Max built **circuit 3** using a battery, two bulbs and three ammeters.



circuit 3

The current reading on ammeter A₁ was 0.8 amps. What would be the reading on ammeters A₂ and A₃?

Place **one** tick in the table by the correct pair of readings.

reading on ammeter A ₂ (amps)	reading on ammeter A ₃ (amps)	correct pair of readings
0.8	0.8	
0.8	0.4	
0.4	0.8	
0.4	0.4	

(1 mark)
(Total 8 marks)

6. Some pupils made an electric cell using two different metals and a lemon.

They put strips of copper and zinc into a lemon and connected them to the terminals of an electric clock.



- a. Look at the photograph.

What evidence is there that they have made an electric cell?

(1 mark)

- b. The pupils had pieces of copper, zinc, iron and magnesium and some lemons.

They wanted to find out which pair of metals made the cell with the biggest voltage. What equipment should they use to measure the voltage of their cells?

(1 mark)

- c. In their investigation they used different pairs of metals. Give one factor that they should keep the same.

(1 mark)

- d. The pupils measured the voltage produced by different pairs of metals. Their results are recorded below.

voltage produced by each pair of metals (volts)				
	magnesium	zinc	iron	copper
copper	1.7	0.9	0.8	0
iron	1.3	0.1	0	-
zinc	0.8	0	-	-
magnesium	0	-	-	-

Which pair of metals made the cell with the biggest voltage?

_____ and _____

(1 mark)

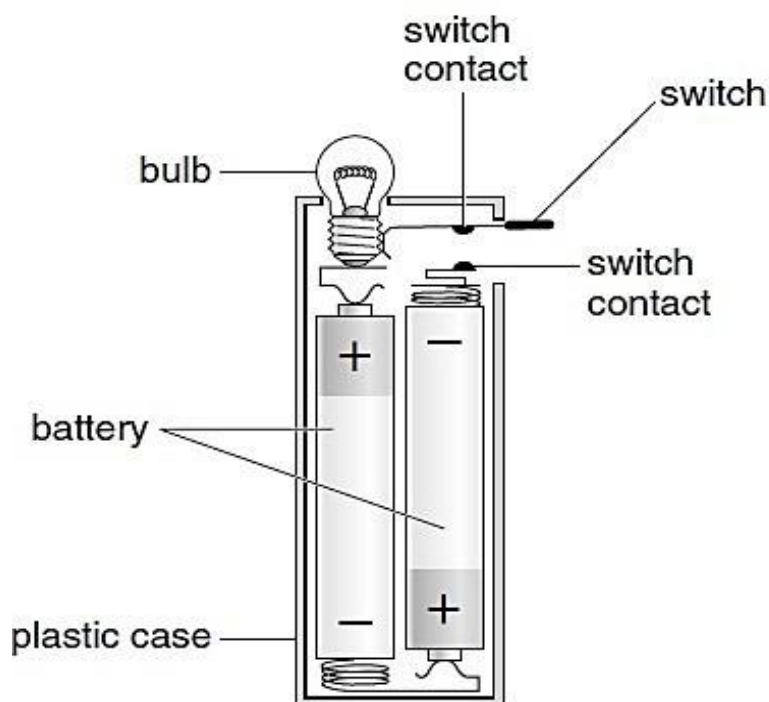
- e. Look at the results in the table above.

Why should the pupils **not** use pairs of the same type of metal for the clock?

(1 mark)

(Total 5 marks)

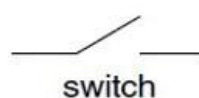
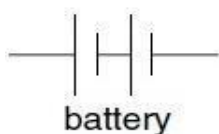
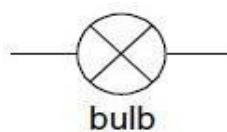
7. The drawing below shows the parts of a torch.



a. i. Paul closed the switch. Why did this turn on the torch?

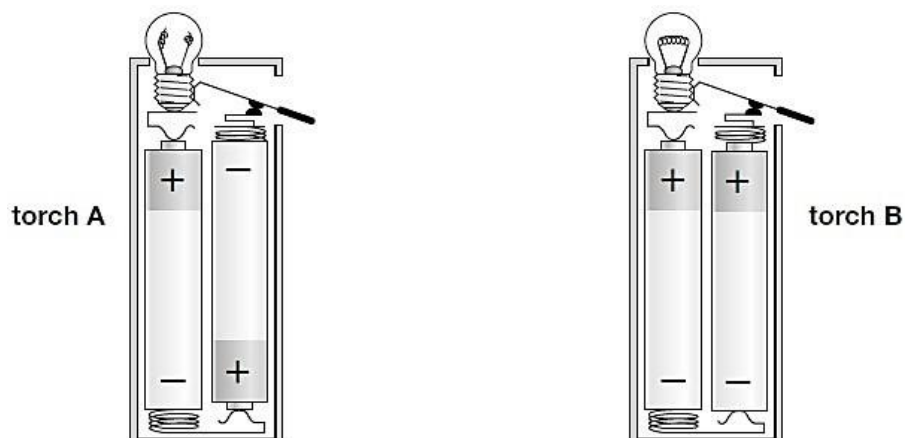
(1 mark)

ii. The diagrams below show symbols for a battery, a bulb and a switch. Connect the symbols to make a series circuit for the torch.



(1 mark)

- b. The drawings below show two other torches. In both torches, the bulbs will **not** light even when Paul closes the switches



Look carefully at the drawings.

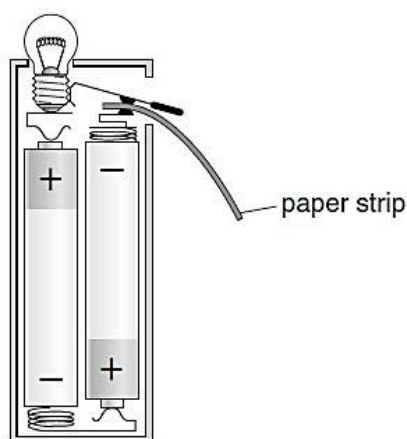
- i. Why is the circuit of torch A **not** complete?

(1 mark)

- ii. What could you do to torch B to get the bulb to light?

(1 mark)

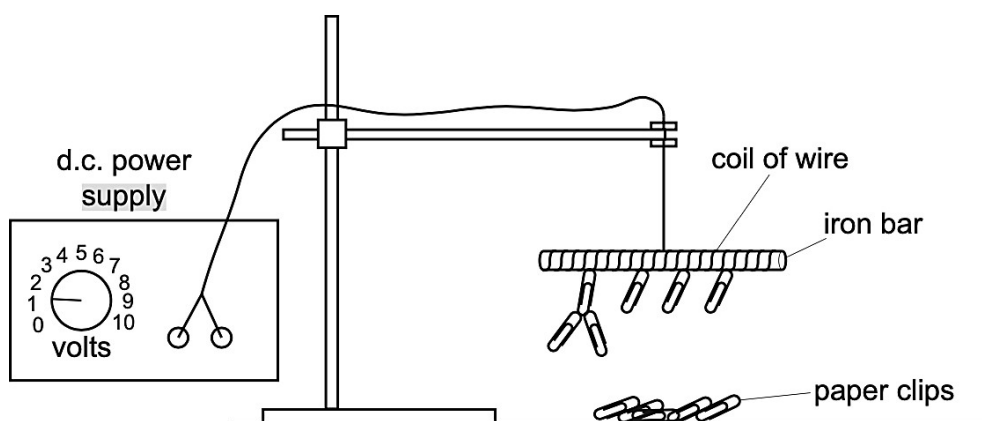
- c. When Paul bought his torch there was a paper strip between the contacts of the switch as shown below.



Paul had to remove the paper strip before he could turn the torch on. Give the reason for this.

(1 mark)
(Total 5 marks)

8. Foram sets up the equipment as shown in the diagram.

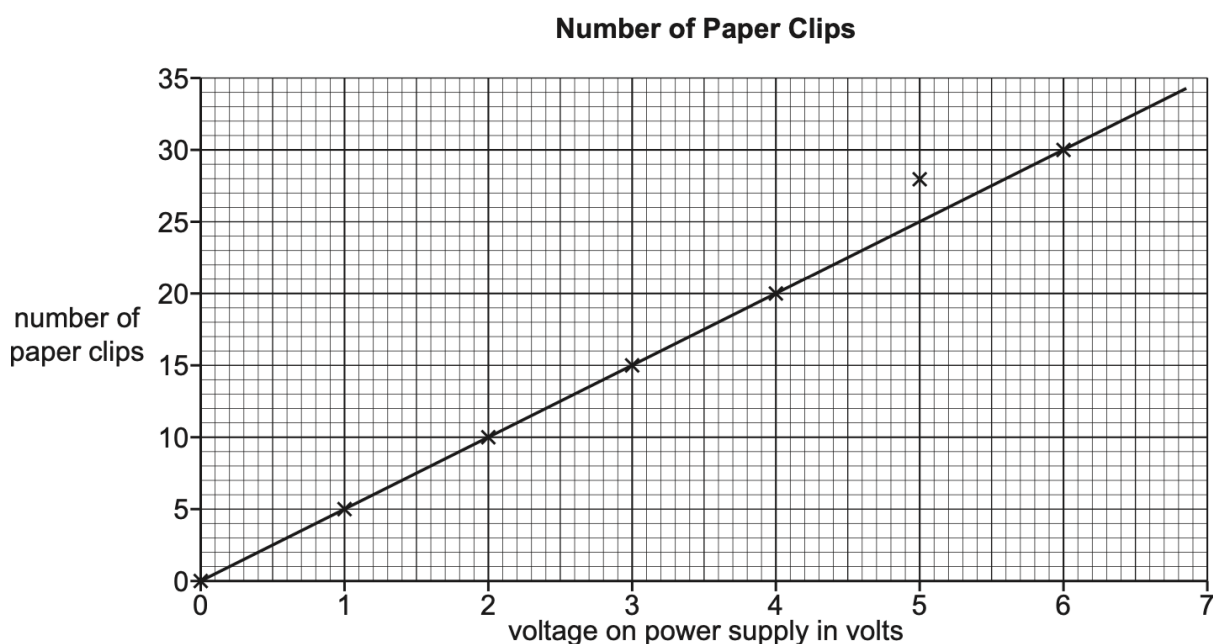


The iron bar attracts steel paper clips when the power supply is switched on. The iron bar and the coil of wire is an electromagnet.

Foram changes the voltage on the power supply.

He records the number of paper clips attracted each time.

a. Here are Foram's results.



On the graph, circle the result that is an anomaly

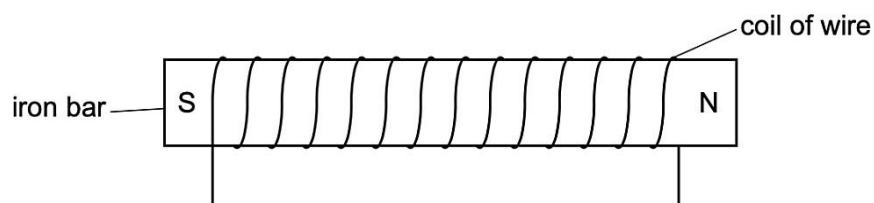
(1 mark)

- b. Foram repeats this experiment using new paper clips made from a different metal.

The equipment was working but did not pick up any of the new paper clips.
Suggest **one** reason for this.

(1 mark)

- c. Foram discovers that the electromagnet creates a magnetic field like a bar magnet. Draw the magnetic field pattern for this electromagnet on the diagram below. Include:
- at least **two** field lines
 - a direction arrow on at least **one** field line.



(2 marks)

(Total 4 marks)

TOTAL FOR ASSIGNMENT 40 MARKS